VANISHING OF HOM INTO A COMPLEX SUPPORTED BELOW THE PROJECTIVE AMPLITUDE, AND THE RESULTING UNIQUE SPLITTING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 19703-19718 (canonical tag 0BCG) and the counted proof is the proof environment at source lines 19720-19741, which immediately follows it and is its own proof.

Editorial note. The four references are two to results of this chapter and two to results of another chapter; the kinds recorded in the stubs are the type words appearing in the labels themselves, as the lane's convention requires, not the names of the environments in which the labels live.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

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Allowed standard prerequisites (editorial).

- An object of projective amplitude in an interval is represented by a complex of projective modules vanishing outside that interval, and a perfect object of tor amplitude there is represented by finite projective modules. invoked at proof lines 19720-19727. Provides the representative of the first object in the vanishing computation, in both the general and the perfect case.
- An object whose cohomology vanishes above a bound is quasi-isomorphic to its truncation below that bound, hence representable by a complex vanishing there. used at proof lines 19727-19731. Puts the second object in the form in which no nonzero map of complexes into it can exist.
- Maps in the derived category from a bounded complex of projectives are maps of complexes, and a distinguished triangle whose first Hom group vanishes splits. used at proof lines 19731-19739. Gives assertion (1), then the splitting of the triangle in assertion (2) and the uniqueness in assertion (3).

Source reference. Original definition 0A5N: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17394–17394. The reference is cited at proof lines 19721-19722 to justify representing the object by a complex of projective modules vanishing outside the interval; the rest of the proof is inside the counted span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0658: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18889–18889. The reference handles the alternative hypothesis that the object is perfect of tor amplitude in the interval, cited at proof lines 19724-19727; it is a pointer and the computation that follows is inside the counted span.

Source reference. Original lemma 064B: sources/stacks-chow/tags.txt, source lines 7860–7860. The reference identifies the Hom group of the derived category with the group of maps of complexes from a bounded complex of projectives, used at proof lines 19732-19734 to conclude the vanishing; the disjointness of supports that makes it apply is inside the counted span.

Source reference. Original lemma 05QT: sources/stacks-chow/tags.txt, source lines 7387–7387. The reference is the criterion saying that a distinguished triangle splits when the relevant Hom group vanishes, applied at proof lines 19736-19738 to get assertion (2); the vanishing it consumes is established inside the counted span.

Lemma 0.2. *Let R be a ring. Let K and L be objects of $D(R)$. Assume L has projective-amplitude in $[a, b]$, for example if L is perfect of tor-amplitude in $[a, b]$.*

1. *If $H^i(K) = 0$ for $i \geq a$, then $\mathrm{Hom}_{D(R)}(L, K) = 0$.*
2. *If $H^i(K) = 0$ for $i \geq a + 1$, then given any distinguished triangle $K \to M \to L \to K[1]$ there is an isomorphism $M \cong K \oplus L$ in $D(R)$ compatible with the maps in the distinguished triangle.*
3. *If $H^i(K) = 0$ for $i \geq a$, then the isomorphism in (2) exists and is unique.*

Proof. The assumption that L has projective-amplitude in $[a, b]$ means we can represent L by a complex $L^\bullet$ of projective R-modules with $L^i = 0$ for $i \notin [a, b]$, see Definition 0A5N. If L is perfect of tor-amplitude in $[a, b]$, then we can represent L by a complex $L^\bullet$ of finite projective R-modules with $L^i = 0$ for $i \notin [a, b]$, see Lemma 0658. If $H^i(K) = 0$ for $i \geq a$, then K is quasi-isomorphic to $\tau_{\leq a-1}K$. Hence we can represent K by a complex $K^\bullet$ of R-modules with $K^i = 0$ for $i \geq a$. Then we obtain

$$\mathrm{Hom}_{D(R)}(L, K) = \mathrm{Hom}_{K(R)}(L^\bullet, K^\bullet) = 0$$

by Derived Categories, Lemma 064B. This proves (1). Under the hypotheses of (2) we see that $\mathrm{Hom}_{D(R)}(L, K[1]) = 0$ by (1), hence the distinguished triangle is split by Derived Categories, Lemma 05QT. The uniqueness of (3) follows from (1). □